Vector functions, gradient and tangent planes

1. Functions, fields and components

A **scalar function** has one number as its value. A **vector function** has a vector as its value. For example,

$$g(t) = t^2, \qquad \mathbf{r}(t) = (t^2, 2t, t^3) = t^2\mathbf{i} + 2t\mathbf{j} + t^3\mathbf{k}.$$

The parameter t may represent time. The vector $\mathbf{r}(t)$ may then give the position of a moving particle.

A **field** assigns a value to every point in a region of space. A temperature $T(x, y, z)$ is a scalar field. A fluid velocity $\mathbf{F}(x, y, z) = (F_1, F_2, F_3)$ is a vector field. Each component F_j can depend on all three coordinates. The **domain** is the set of allowed inputs; all polynomial examples below are defined for every real input.

Vector reminders

For $\mathbf{a} = (a_1, a_2, a_3)$ and $\mathbf{b} = (b_1, b_2, b_3)$,

$$|\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}, \qquad \mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3.$$

For nonzero vectors, $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta$. Thus a zero dot product means perpendicular directions. A unit vector has length 1.

2. Vector limits and derivatives

Limits and continuity work component by component

For $\mathbf{r}(t) = (r_1(t), r_2(t), r_3(t))$,

$$\lim_{t \to t_0} \mathbf{r}(t) = \left(\lim_{t \to t_0} r_1(t), \lim_{t \to t_0} r_2(t), \lim_{t \to t_0} r_3(t) \right)$$

when the three finite limits exist. This says that the distance from $\mathbf{r}(t)$ to the limiting vector tends to zero. The function is **continuous at** t_0 when this limit equals $\mathbf{r}(t_0)$; equivalently, all its components are continuous there.

Derivative and motion

When the limit exists,

$$\mathbf{r}'(t) = \lim_{h \to 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h} = (r_1'(t), r_2'(t), r_3'(t)).$$

For a position vector $\mathbf{r}(t)$:

$$\text{velocity } \mathbf{v} = \mathbf{r}', \qquad \text{acceleration } \mathbf{a} = \mathbf{r}'', \qquad \text{speed } = |\mathbf{v}|.$$

At a point where $\mathbf{r}' \neq \mathbf{0}$, the derivative gives the direction tangent to the path, and the unit tangent is $\mathbf{r}'/|\mathbf{r}'|$.

The component formula follows because subtraction, division by h and the limit all act component by component. The basis vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are fixed, so they are not differentiated.

Example 1. Position, velocity and speed

For $\mathbf{r}(t) = (t^2, 2t, t^3)$, find the position, velocity, acceleration and speed at $t = 1$.

Solution. Differentiate first, then substitute:

$$\begin{aligned} \mathbf{r}(1) &= (1,2,1), \\ \mathbf{r}'(t) &= (2t, 2, 3t^2), \qquad \mathbf{v}(1) = (2,2,3), \\ \mathbf{r}''(t) &= (2,0,6t), \qquad \mathbf{a}(1) = (2,0,6), \\ |\mathbf{v}(1)| &= \sqrt{2^2+2^2+3^2} = \boxed{\sqrt{17}}. \end{aligned}$$

The unit tangent is $(2,2,3)/\sqrt{17}$. If position is measured in metres and t in seconds, the velocity components are in $\mathrm{m\,s^{-1}}$, acceleration in $\mathrm{m\,s^{-2}}$, and speed in $\mathrm{m\,s^{-1}}$.

3. Differentiation rules and vector partial derivatives

Rules for differentiable vector functions

Let $\mathbf{u}(t), \mathbf{v}(t)$ be differentiable, let $\phi(t)$ be a differentiable scalar function, and let c be constant.

$$\begin{aligned} (c\mathbf{u})' &= c\mathbf{u}', & (\mathbf{u}+\mathbf{v})' &= \mathbf{u}'+\mathbf{v}', \\ (\phi\mathbf{u})' &= \phi'\mathbf{u} + \phi\mathbf{u}', \\ (\mathbf{u}\cdot\mathbf{v})' &= \mathbf{u}'\cdot\mathbf{v} + \mathbf{u}\cdot\mathbf{v}', \\ (\mathbf{u}\times\mathbf{v})' &= \mathbf{u}'\times\mathbf{v} + \mathbf{u}\times\mathbf{v}'. \end{aligned}$$

A dot product gives a scalar; a cross product gives a vector. In the cross-product rule keep the order of the factors: generally $\mathbf{v}\times\mathbf{u} = -\mathbf{u}\times\mathbf{v}$.

These are the usual product rules applied to components. For example, differentiating $u_1v_1 + u_2v_2 + u_3v_3$ gives exactly the two dot products in the rule. Complete dot- and cross-product calculations appear in Example 4.

A partial derivative of a whole vector

For a vector field $\mathbf{F} = (F_1, F_2, F_3)$, hold the other variables fixed and differentiate *every* component:

$$\boxed{\mathbf{F}_x = \frac{\partial \mathbf{F}}{\partial x} = \left(\frac{\partial F_1}{\partial x}, \frac{\partial F_2}{\partial x}, \frac{\partial F_3}{\partial x}\right).}$$

For $\mathbf{F} = (xy,\ y^2+z,\ xz)$, this gives

$$\begin{aligned} \mathbf{F}_x &= (y, 0, z), \\ \mathbf{F}_y &= (x, 2y, 0), \\ \mathbf{F}_z &= (0, 1, x). \end{aligned}$$

In $\mathbf{F}_x$, for example, y and z are constants: $(xy)_x = y$, $(y^2+z)_x = 0$ and $(xz)_x = z$. Second partial derivatives also act componentwise; here $\mathbf{F}_{xx} = (0,0,0)$ and $\mathbf{F}_{xy} = (1,0,0) = \mathbf{F}_{yx}$. Continuous second partial derivatives are sufficient for equality of the mixed partials.

4. Gradient and change in a chosen direction

Gradient and directional derivative

Suppose a scalar function $f(x, y, z)$ has continuous first partial derivatives near P. The operator $\nabla = \mathbf{i}\,\partial_x + \mathbf{j}\,\partial_y + \mathbf{k}\,\partial_z$ is read *nabla*, where ∂_x means $\partial/\partial x$.

$$\nabla f = (f_x, f_y, f_z)$$

is the **gradient**: three coordinate rates gathered into one vector. For a *unit* vector $\mathbf{u}$,

$$D_{\mathbf{u}} f(P) = \lim_{s \to 0} \frac{f(P + s\mathbf{u}) - f(P)}{s} = \boxed{\nabla f(P) \cdot \mathbf{u}}.$$

This is the rate of change per unit distance in the chosen direction.

Why the dot-product formula works

Write $P = (x_0, y_0, z_0)$ and $\mathbf{u} = (u_1, u_2, u_3)$. On the line

$$x = x_0 + su_1, \qquad y = y_0 + su_2, \qquad z = z_0 + su_3,$$

the chain rule gives

$$\left.\frac{\mathrm{d}f}{\mathrm{d}s}\right|_{s=0} = f_x(P)u_1 + f_y(P)u_2 + f_z(P)u_3 = \nabla f(P) \cdot \mathbf{u}.$$

Because $|\mathbf{u}| = 1$, the distance from P is $|s|$. Without normalising the direction, the derivative would be with respect to a differently scaled parameter.

Example 2. Direction towards another point

Let $f = x^2 + xy + z^2$. Find its rate of change at $P = (1, 1, 1)$ towards $Q = (3, 0, 3)$, and its maximum directional rate at P.

1. Find the gradient at the starting point

$$f_x = 2x + y, \qquad f_y = x, \qquad f_z = 2z,$$

so

$$\nabla f = (2x + y, x, 2z), \qquad \nabla f(P) = (3, 1, 2).$$

2. Turn the displacement into a unit direction

$$Q - P = (3 - 1, 0 - 1, 3 - 1) = (2, -1, 2), \qquad |Q - P| = \sqrt{4 + 1 + 4} = 3.$$

Thus $\mathbf{u} = (2, -1, 2)/3$, whose length is 1.

3. Take the dot product

$$D_{\mathbf{u}} f(P) = (3, 1, 2) \cdot \frac{(2, -1, 2)}{3} = \frac{6 - 1 + 4}{3} = \boxed{3}.$$

The positive sign means that f initially increases in this direction. The answer is a local rate at P, not the average change from P to Q.

4. Find the largest possible rate

For any unit direction $\mathbf{e}$, the dot-product identity gives

$$D_{\mathbf{e}} f(P) = |\nabla f(P)| \cos\theta, \qquad |\nabla f(P)| = \sqrt{3^2 + 1^2 + 2^2} = \sqrt{14}.$$

Here θ is the angle between $\mathbf{e}$ and the gradient. Since $-1 \le \cos\theta \le 1$,

$$\boxed{\text{maximum rate } = \sqrt{14} \quad \text{in direction } \frac{(3, 1, 2)}{\sqrt{14}}.}$$

The minimum is $-\sqrt{14}$ in the opposite direction. Directions perpendicular to the gradient give zero first-order change. The result $3 < \sqrt{14}$ checks our chosen-direction answer. If the gradient is zero, all directional derivatives given by this formula are zero; there is no unique direction of fastest first-order increase. For example, at the origin $h(x, y, z) = x^2 + y^2 + z^2$ has gradient zero. Along any unit direction, $h(s\mathbf{u}) = s^2$ still changes for $s \neq 0$, but its derivative at $s = 0$ is zero. This is a first-order statement.

A temperature interpretation

If a local temperature model is $T = 25 + f$ in degrees Celsius, with numerical coordinates in metres, then $T(P) = 28\,^\circ\mathrm{C}$. The derivative towards Q is $3\,^\circ\mathrm{C\,m^{-1}}$. For a small positive distance s metres, the first-order estimate is a temperature rise of $3s\,^\circ\mathrm{C}$.

5. Level surfaces, normals and tangent planes

A **level surface** $f(x, y, z) = c$ contains the points where the scalar field has the same value c. A **normal vector** is perpendicular to the tangent plane at the point.

Why the gradient is normal

Let a differentiable curve $\mathbf{r}(t)$ lie on $f = c$ and pass through P at $t = t_0$. Along this curve the scalar value is constant, so

$$f(\mathbf{r}(t)) = c \quad \Longrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 0.$$

At P, this says $\nabla f(P) \cdot \mathbf{r}'(t_0) = 0$. Thus the gradient is perpendicular to each tangent direction of the surface. If the first partial derivatives are continuous and $\nabla f(P) \neq \mathbf{0}$, the surface is regular near P and this nonzero vector is a normal.

The tangent plane from a point and a normal

For $P = (x_0, y_0, z_0)$ and $\mathbf{n} = \nabla f(P) \neq \mathbf{0}$, a point $X = (x, y, z)$ lies in the tangent plane exactly when

$$\mathbf{n} \cdot (X - P) = 0.$$

Therefore

$$\boxed{f_x(P)(x - x_0) + f_y(P)(y - y_0) + f_z(P)(z - z_0) = 0.}$$

Use the gradient *at* P; the coefficients of the plane are constants.

Example 3. Reuse the gradient

Find the tangent plane and a normal line to $x^2 + xy + z^2 = 3$ at $P = (1, 1, 1)$.

Solution. First verify the point: $1^2 + 1 \cdot 1 + 1^2 = 3$, so P is on the surface. The defining polynomial has continuous first partial derivatives, and Example 2 gives the nonzero normal $\mathbf{n} = (3, 1, 2)$. Hence

$$3(x - 1) + (y - 1) + 2(z - 1) = 0.$$

Expanding the three brackets gives $3x - 3 + y - 1 + 2z - 2 = 0$, or

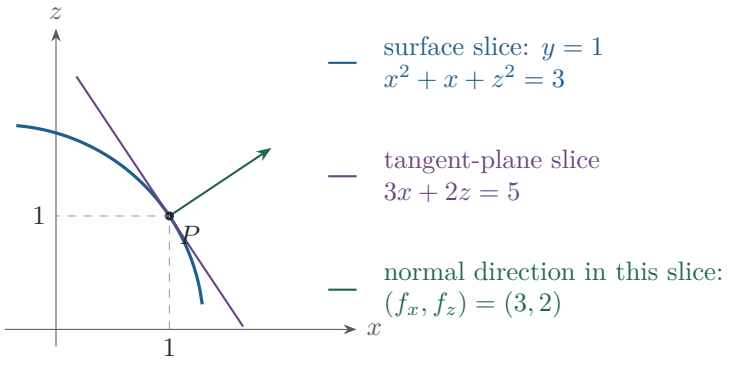
$$\boxed{3x + y + 2z = 6}.$$

It passes through P because $3 + 1 + 2 = 6$, and its normal is $(3, 1, 2)$ as required.

The **normal line** passes through P in direction $\mathbf{n}$:

$$\boxed{(x, y, z) = (1, 1, 1) + \lambda(3, 1, 2), \qquad \lambda \in \mathbb{R}.}$$

Equivalently, $x = 1 + 3\lambda$, $y = 1 + \lambda$, $z = 1 + 2\lambda$. The two unit normals are $\pm(3, 1, 2)/\sqrt{14}$.



Cutting at $y = 1$ gives the blue surface curve and the purple tangent line. They touch at P. The normal $(3, 2)$ belongs to this xz-slice; the full three-dimensional normal remains $(3, 1, 2)$.
For the temperature model $T = 25 + f$, this is the surface $T = 28\,^\circ\text{C}$. The tangent plane gives its local flat approximation. Moving in a tangent direction gives zero first-order temperature change.

6. Understanding checks

1. **If $\mathbf{F} = (xy, y^2 + z, xz)$, is $\mathbf{F}_x = (y, 0, 0)$?**
   No. The third component xz also depends on x, so $\mathbf{F}_x = (y, 0, z)$.
2. **Is $\nabla f(P) \cdot (Q - P)$ always a rate per unit distance?**
   No. First use $(Q - P)/|Q - P|$; the direction must have length 1.
3. **Why is the plane $3x + y + 2z = 0$ wrong in Example 3?**
   It has the correct normal but does not contain P. The correct right-hand side is $3(1) + 1 + 2(1) = 6$.

7. Worked product-rule checks

Example 4. Dot and cross products

Let $\mathbf{u}(t) = (t, t^2, 1)$ and $\mathbf{v}(t) = (1, 2, t)$. Differentiate their dot and cross products, and check the product rules.

$$\mathbf{u}' = (1, 2t, 0), \qquad \mathbf{v}' = (0, 0, 1).$$

Dot product: a scalar

Direct multiplication gives

$$\mathbf{u} \cdot \mathbf{v} = t + 2t^2 + t = 2t + 2t^2 \quad \Longrightarrow \quad (\mathbf{u} \cdot \mathbf{v})' = 2 + 4t.$$

The product rule gives the same answer:

$$\mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}' = (1 + 4t + 0) + (0 + 0 + 1) = \boxed{2 + 4t}.$$

Cross product: a vector

In a right-handed Cartesian system,

$$\mathbf{a} \times \mathbf{b} = (a_2 b_3 - a_3 b_2,\ a_3 b_1 - a_1 b_3,\ a_1 b_2 - a_2 b_1).$$

Therefore

$$\mathbf{u} \times \mathbf{v} = (t^2 t - 1 \cdot 2,\ 1 \cdot 1 - t \cdot t,\ t \cdot 2 - t^2 \cdot 1) = (t^3 - 2, 1 - t^2, 2t - t^2).$$

Differentiation gives $(3t^2, -2t, 2 - 2t)$. Now compute the two terms in the rule:

$$\begin{aligned}
\mathbf{u}' \times \mathbf{v} &= (2t \cdot t - 0 \cdot 2,\ 0 \cdot 1 - 1 \cdot t,\ 1 \cdot 2 - 2t \cdot 1) \\
&= (2t^2, -t, 2 - 2t), \\
\mathbf{u} \times \mathbf{v}' &= (t^2 \cdot 1 - 1 \cdot 0,\ 1 \cdot 0 - t \cdot 1,\ t \cdot 0 - t^2 \cdot 0) \\
&= (t^2, -t, 0).
\end{aligned}$$

Adding gives

$$\mathbf{u}' \times \mathbf{v} + \mathbf{u} \times \mathbf{v}' = \boxed{(3t^2, -2t, 2 - 2t)}.$$

Reversing one cross product would change its sign and spoil the rule.

A useful consequence: constant length

If a differentiable vector $\mathbf{w}(t)$ has constant length c, then

$$\mathbf{w} \cdot \mathbf{w} = c^2 \quad \Longrightarrow \quad \mathbf{w}' \cdot \mathbf{w} + \mathbf{w} \cdot \mathbf{w}' = 0 \quad \Longrightarrow \quad \mathbf{w} \cdot \mathbf{w}' = 0.$$

A nonzero $\mathbf{w}'$ is perpendicular to $\mathbf{w}$; a vector can turn while keeping its length. If $\mathbf{w}' = \mathbf{0}$, there is no derivative direction.

The same rule with three factors

For differentiable vector functions, the scalar triple product $[\mathbf{u}, \mathbf{v}, \mathbf{w}] = \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t}[\mathbf{u}, \mathbf{v}, \mathbf{w}] = [\mathbf{u}', \mathbf{v}, \mathbf{w}] + [\mathbf{u}, \mathbf{v}', \mathbf{w}] + [\mathbf{u}, \mathbf{v}, \mathbf{w}'].$$

Apply the dot-product rule, then the cross-product rule, retaining the factor order.

8. Alternative checks and delicate points

Check the directional derivative directly

For Example 2, $P + s\mathbf{u} = (1 + 2s/3,\ 1 - s/3,\ 1 + 2s/3)$. Substitution into $f = x^2 + xy + z^2$ gives

$$\begin{aligned} f(P + s\mathbf{u}) &= 2\left(1 + \frac{2s}{3}\right)^2 + \left(1 + \frac{2s}{3}\right)\left(1 - \frac{s}{3}\right) \\ &= 2\left(1 + \frac{4s}{3} + \frac{4s^2}{9}\right) + 1 + \frac{s}{3} - \frac{2s^2}{9} \\ &= 3 + 3s + \frac{2}{3}s^2. \end{aligned}$$

Its derivative at $s = 0$ is 3, agreeing with the gradient method. For a finite displacement there is also a quadratic term; the directional derivative is the first-order rate.

The familiar graph formula gives the same plane

Near $P = (1, 1, 1)$ the positive branch of the level surface is

$$z = g(x, y) = \sqrt{3 - x^2 - xy}.$$

The expression under the root equals 1 at P, so this branch is differentiable nearby. The chain rule gives

$$g_x = -\frac{2x + y}{2\sqrt{3 - x^2 - xy}}, \qquad g_y = -\frac{x}{2\sqrt{3 - x^2 - xy}}.$$

At $(x, y) = (1, 1)$ these are $-3/2$ and $-1/2$. The graph tangent-plane formula therefore gives

$$z - 1 = -\frac{3}{2}(x - 1) - \frac{1}{2}(y - 1).$$

Multiplying by 2 and rearranging gives $3x + y + 2z = 6$. For any graph $z = g(x, y)$, the defining function $F = g(x, y) - z$ has normal $(g_x, g_y, -1)$ and gives this same graph formula.

Why the nonzero-gradient condition matters

A zero gradient makes the plane equation become $0 = 0$, so this method fails. The surface may still have a tangent plane: $f = z^2$ has zero gradient on $f = 0$, although that level set is the plane $z = 0$. Here use the regular defining function $F = z$.